

Oswald Gyabaah

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PROFESSIONAL SUMMARY

Engineering leader and architect with 6 years building and operating fintech-grade systems end to end. Founding CTO of MyHealthCop, where I built a 12-person engineering organization and the full platform under it. Founded Guaranty Trust Bank Ghana's DevOps practice from scratch, and now run the on-prem core banking platform behind 1.5M+ daily financial transactions. I architect scalable microservices and distributed systems, set standards for code quality and security, and mentor teams through full delivery lifecycles. Daily work in Python, React, TypeScript, and PostgreSQL.

CORE COMPETENCIES

Engineering Leadership & Mentorship

Software Architecture

Microservices & Distributed Systems

Python (Django, FastAPI)

React & TypeScript

PostgreSQL & MySQL

CI/CD & Automated Testing

Fintech & Payments Systems

WORK EXPERIENCE

Guaranty Trust Bank Ghana LTD

May 2025 - Present

Core Banking Services Lead - Accra, Ghana

- Own the on-prem Finacle core banking deployment sustaining **1.5M+ daily financial transactions**, managing deployments, patching, and incident response across 800+ production pods on Tanzu Kubernetes Grid.
- Architected integrations between Finacle core banking and third-party systems including transaction portals and internal tools, using REST APIs and service-layer patterns.
- Led a 5-person QA team through the bank's Basis-to-Finacle migration cutover, designing and executing 10,000+ test cases across 13 core banking modules and 33 internal applications.
- Built a bank-wide OTP service in Python (Django) and Next.js, integrating SMS and email gateways to secure core banking access for 600+ internal users, with structured audit logging to Elasticsearch.
- Built a national ID verification (KYC) service processing and storing ID documents in self-hosted MinIO, with automated verification and customer record updates.
- Extended the bank's observability into core banking with Finacle-specific dashboards and threshold-based alerting for proactive scaling.

Guaranty Trust Bank Ghana LTD

Apr 2024 - Present

DevOps Engineering Lead - Accra, Ghana

- Founded and led the bank's DevOps team of 4 engineers, fostering a culture of code quality and continuous improvement; replaced manual code handoffs with enforced Git workflows and containerized Kubernetes deployments.
- Implemented CI/CD automation that cut deployment time ~90% and increased release velocity ~5x; introduced GitOps with FluxCD, and integrated Veracode and Trivy scanning for automated security analysis.
- Architected high-availability Kubernetes clusters across multiple on-prem environments running internal applications handling 3M+ daily requests, with 99.97% measured uptime; implemented auto-scaling and self-healing.
- Automated 95% of infrastructure configuration management with Ansible, Python, and Bash, turning multi-day manual processes into reproducible, version-controlled deployments.
- Introduced a unified observability stack (Prometheus, Grafana, Elastic Stack, Sentry) for centralized logging, tracing, and real-time alerting, reducing mean time to resolution on production incidents.

Chief Technical Officer (Founding) - Accra, Ghana

- Led a **12-person cross-functional engineering team** (frontend, backend, mobile, product, design, QA) through the full project lifecycle: requirements gathering, sprint planning, stakeholder management, and budgeting under Agile/Scrum.
- Architected the myhealthcop.com platform end to end across system design, database modeling, and API development, delivering a React admin web dashboard and two Flutter mobile apps.
- Designed secure, cost-efficient AWS infrastructure running containerized microservices on EKS and ECS alongside Lambda for event-driven workloads, with autoscaling, load balancing, and inter-service communication.
- Built a high-throughput background processing system handling 1M+ daily jobs across payments, messaging, and notifications using SQS/SNS, with SMS, email, and payment provider integrations including multi-provider failover.
- Led performance optimizations that reduced peak request latency by **90%** through Redis caching, query tuning, and PostgreSQL indexing.
- Designed a disaster recovery strategy mirroring the AWS production environment on GCP for rapid failover, with secure VPCs, IAM policies, and dual-environment CI/CD.

PROJECTS

Digital-Recon - Transaction Reconciliation Engine Fintech

Founder. Real-time transaction reconciliation engine for financial institutions, detecting discrepancies across heterogeneous data sources - relational databases, APIs, and Elasticsearch - in real time. Architected an event-driven processing layer using Kafka for upstream transaction ingestion and Celery for scheduled and on-demand reconciliation jobs.

Stack: React, TypeScript, Python, Celery, Kafka, Elasticsearch, PostgreSQL, Docker

AI Agents for Engineering Operations Python & AI/ML

Personal R&D. Building specialized AI agents for QA automation, code review, and sysadmin operations, running against live Linux production servers. Prototyping agent-driven tooling using RAG, tool-use, and multi-agent orchestration patterns to raise the consistency of code review and quality checks.

Stack: Python, Anthropic API, OpenAI API, RAG, tool-use, AWS Lambda, EventBridge

EDUCATION

BSc Computer Engineering - Ashesi University

Sep 2016 - May 2020

Berekuso, Ghana

CERTIFICATIONS

AI Engineering Specialization - ByteByteAI	Mar 2026
Oracle Certified Architect Associate - Oracle	Aug 2025
Oracle Certified DevOps Professional - Oracle	Aug 2025
AWS Certified Cloud Practitioner - AWS	May 2024
Django Web Framework - Coursera	Dec 2022

SKILLS

Backend & Architecture: Python (Django, FastAPI, Celery), Node.js (Express), REST APIs, GraphQL, WebSockets, microservices, service-oriented architecture, distributed systems

Frontend: React.js, Next.js, TypeScript, JavaScript, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Firebase/Firestore - performance tuning, schema design, indexing

Leadership & Process: Engineering team leadership, mentorship, code review, architectural audits, Agile/Scrum, sprint planning, stakeholder management, full project lifecycle

CI/CD & Testing: GitHub Actions, Azure DevOps, FluxCD, ArgoCD, Terraform, Ansible; Jest, React Testing Library, Cypress, Playwright, pytest

Cloud & Infrastructure: AWS (EKS, ECS, Lambda, SQS/SNS), GCP (GKE), Kubernetes, Docker, Helm

Observability & Security: Prometheus, Grafana, ELK Stack, Sentry; Veracode, Trivy, KYC/identity verification